Department of Information Science & Engineering

Artificial Intelligence and Machine Learning

IS353IA

PROJECT SYNOPSIS

Lab – Part B

Title of Project:	
Student Name(s):	
USN / Reg. No.:	
Semester and Section:	
Lab Batch:	
Date of Submission:	/

Academic Year:

1. Abstract

Guidelines:

Write 200–300 words.

Cover: problem being solved, proposed approach, methodology, expected outcome.

Must be self-contained — a reader unfamiliar with your work should understand it.

No citations or references in the abstract.

Avoid acronyms unless defined on first use.

[Write a concise overview of the proposed project here. Summarize the problem, the proposed solution, the methodology to be followed, tools/technologies involved, and the expected outcomes or contributions. Do NOT copy from existing papers. Keep it original and specific to your work.]

Keywords: *[5–7 relevant keywords separated by commas; e.g., Federated Learning, Brain Tumour Detection, Deep Learning, MRI Segmentation, Privacy-Preserving AI]*

2. Introduction

Guideline:

Length: 300–500 words.

Provide background context for the problem domain.

State what is known, what is unknown or unsolved, and why this project matters.

End with a clear problem statement (1–3 sentences).

Cite recent literature (last 5 years preferred) to support claims.

2.1 Background

[Introduce the broad domain (e.g., Healthcare AI, Cybersecurity, NLP). Explain the current state of the field and key challenges. Use recent data, statistics, or industry context to justify relevance.]

2.2 Problem Statement

[State the precise problem your project addresses. Be specific — avoid vague statements. Example: 'Existing intrusion detection systems fail to generalise across heterogeneous network environments due to the lack of federated training, resulting in high false-positive rates in cross-domain deployments.']

2.3 Motivation

[Explain why this problem is important to solve NOW. What are the consequences if it remains unsolved? Who is affected?]

3. Literature Survey / Review of Related Work

Guideline:

Minimum 8–10 references reviewed; summarise key works in 2–3 sentences each.

Identify research gaps from existing work.

Organise by sub-theme, not chronologically.

Every claim must be backed by a citation [Author, Year].

End with a Research Gap Summary table (provided below as a template).

[Discuss existing work in your domain. Group papers by theme. Example sub-sections below:]

3.1 [Sub-theme 1 — e.g., Deep Learning for Medical Imaging]

[Summarise 2–3 papers. Example: Author et al. [Year] proposed a CNN-based model for X achieving Y% accuracy on the Z dataset. However, their approach does not account for ...]

3.2 [Sub-theme 2 — e.g., Privacy-Preserving Techniques]

[Continue with next cluster of related work...]

3.3 Research Gap Summary

Reference	Key Contribution	Limitation / Research Gap
[Author, Year]	[What the paper proposed]	[What it did NOT address — which your project will]
[Author, Year]	[What the paper proposed]	[What it did NOT address]
[Author, Year]	[What the paper proposed]	[What it did NOT address]

4. Objectives

Guideline:

List 3–4 specific, measurable, achievable objectives.

Each objective should start with an action verb (Design, Develop, Implement, Evaluate, Compare, Optimise).

Objectives should collectively address the research gap identified.

Do NOT write vague objectives like 'To study the literature'.

The primary objectives of this project are:

- To [Action verb + specific goal — e.g., design a federated learning framework for decentralised brain tumour classification].
- To [Action verb + specific goal — e.g., implement differential privacy mechanisms to ensure data confidentiality across participating institutions].
- To [Action verb + specific goal — e.g., evaluate the proposed model on benchmark datasets (BraTS 2024, TCGA) and compare against state-of-the-art centralised baselines].
- To [Action verb + specific goal — e.g., optimise communication efficiency using gradient compression techniques].

5. Proposed Methodology

Guideline:

This is the most critical section. Describe your approach clearly and technically.

Include: system architecture / block diagram description, algorithms or models to be used, dataset(s), tools and technologies, and implementation steps.

If using ML/DL: specify model architecture, loss functions, optimisers, hyperparameters (at least planned).

Distinguish clearly what is novel versus what is adopted from prior work.

Add a figure placeholder for system architecture diagram.

5.1 System Architecture Overview

[Describe the high-level architecture. Insert a figure here showing the system pipeline or block diagram.]

[INSERT SYSTEM ARCHITECTURE DIAGRAM / BLOCK DIAGRAM HERE]

Figure 1: Proposed System Architecture

5.2 Dataset Description

[Name the dataset(s) to be used. Provide source (URL or citation), size, format, class distribution, and any pre-processing steps planned.]

5.3 Proposed Algorithm / Model

[Describe the core algorithm, model, or technique. If ML/DL: architecture layers, activation functions, loss function, optimiser. If a novel algorithm: provide pseudocode or numbered steps.]

5.4 Implementation Plan (Phase-wise)

Phase	Task to be completed	Activities involved	Duration
I	Literature Review & Problem Definition	Survey existing work, finalise problem statement, define scope	Weeks
II	Dataset Acquisition & Pre-processing	Collect/download dataset, data cleaning, augmentation, EDA	Weeks
III	System Design & Architecture	Design system block diagram, select models, prepare environment	Weeks
IV	Implementation & Training	Code development, model training, hyperparameter tuning	Weeks
V	Evaluation & Testing	Performance metrics, ablation study, comparison with baselines	Weeks
VI	Documentation & Reporting	Final report, paper draft, presentation preparation	Weeks

5.5 Tools & Technologies

Category	Tools / Technologies / Versions
Programming Language	
Frameworks / Libraries	
Development Environment	*[Jupyter Notebook / VS Code / Google Colab / PyCharm]*
Database / Storage	*[MySQL / MongoDB / PostgreSQL / SQLite / HDF5 / AWS S3]*
Hardware / Platform	*[GPU: NVIDIA RTX 3090 / Google Colab T4 / DGX Server; RAM: 16 GB]*
Version Control	*[Git / GitHub / GitLab]*
Other Tools	*[Docker / Kubernetes / Weights & Biases / MATLAB / Tableau]*

6. Expected Outcomes & Deliverables

Guideline:

Be specific and measurable. Avoid vague statements like 'good accuracy'.

Mention quantitative targets where possible (e.g., >90% F1-score on ABCD dataset).

List deliverables: software prototype, dataset, model weights, paper, etc.

Upon successful completion of this project, the following outcomes are expected:

6.1 Technical Outcomes

- [e.g., A trained and validated model achieving >92% classification accuracy on the benchmark dataset]
- [e.g., A fully functional software prototype / web application / API]
- [e.g., A comparative evaluation report against 3 state-of-the-art baseline methods]
- [e.g., Reduction of communication overhead by >30% using gradient compression]

6.2 Research Outcomes

- [e.g., Publication of findings in a peer-reviewed Q1/Q2 Scopus/SCI indexed journal or IEEE/Springer conference]
- [e.g., A novel algorithm / framework with measurable improvements over existing work]

6.3 Deliverables Checklist

Deliverable	Target Timeline
Synopsis / Proposal Document	
Literature Review Report	
Working Prototype / Baseline Model	
Final Trained Model + Code Repository	
Evaluation Report & Results	
Journal / Conference Paper Draft	
Final Project Report & Presentation	Week 18

7. Evaluation Metrics & Validation Plan

Guideline:

Specify how you will measure the success of your project.

List relevant metrics for your domain (accuracy, F1, BLEU, FID, etc.).

Mention train/validation/test split, cross-validation strategy, or benchmark comparison approach.

Include ethical considerations if using human data.

[Describe metrics relevant to your task:]

- [Classification tasks: Accuracy, Precision, Recall, F1-Score, AUC-ROC, Confusion Matrix]
- [Segmentation tasks: Dice Coefficient, IoU (Jaccard Index), Hausdorff Distance]
- [NLP tasks: BLEU, ROUGE, METEOR, Perplexity, BERTScore]

- [System performance: Latency (ms), Throughput (req/s), Memory footprint (MB)]
- [Security/Privacy: Differential Privacy ε-budget, Re-identification risk]

Validation Strategy:

- [e.g., 80/10/10 train-validation-test split with stratified sampling]
- [e.g., 5-fold cross-validation to ensure statistical robustness]
- [e.g., Comparison against 3 baseline methods on the same dataset under identical conditions]

Ethical Considerations:

- [e.g., Data anonymisation / de-identification procedures]
- [e.g., IRB/IEC approval if human subjects or patient data is involved]
- [e.g., Bias analysis across demographic subgroups]

8. References

Guideline:

Minimum 10 references; prefer works from the last 5 years (2020–2025).

Follow IEEE citation format (recommended for CS/Engineering).

Every reference listed must be cited in the text above.

Number references sequentially in the order they first appear in text.

Use a consistent citation manager (Mendeley / Zotero / Google Scholar) to avoid errors.

Examples of IEEE format are provided below — follow the same pattern.

IEEE Format Examples:

[1] A. B. Author and C. D. Author, "Title of journal article," IEEE Trans. Neural Netw. Learn. Syst., vol. 34, no. 5, pp. 1234–1245, May 2023. doi: 10.1109/TNNLS.2022.XXXXXXX

[2] E. F. Author, "Title of conference paper," in Proc. Int. Conf. Machine Learning (ICML), Vienna, Austria, 2024, pp. 567–578.

[3] G. H. Author, Title of Book, 3rd ed. New York, NY: Springer, 2021.

[4] I. J. Author, "Title of thesis," M.Tech. thesis, Dept. Comput. Sci. Eng., Indian Inst. Technol., Bangalore, India, 2023.

[5] K. L. Author. (2024). "Title of online resource." [Online]. Available: https://example.com/resource. [Accessed: 10 Sept. 2025].

[6] [Add your references here following IEEE format]

[7] [Add your references here following IEEE format]
[8] [Add your references here following IEEE format]
[9] [Add your references here following IEEE format]
[10] [Add your references here following IEEE format]

9. Declaration & Signatures

I/We hereby declare that this project synopsis is our original work carried out as part of the the AI and Ml Course . . This work has not been submitted elsewhere for any other degree or examination. All sources of information have been duly cited and acknowledged.

Student Signature:	**Guide Signature:**
_____	_____
[Name & USN]	*[Dr./Prof. Name, Designation]*
Date: _____	*Date: _____*

Appendix: Formatting & Submission Guidelines

A. Document Formatting Requirements

Parameter	Specification
Paper Size	A4 (210 × 297 mm)
Margins	Top: 1 inch | Bottom: 1 inch | Left: 1.25 inch | Right: 1 inch
Font (Body)	Times New Roman 12pt OR Calibri 11pt
Font (Headings)	Same font, Bold: H1 = 14pt, H2 = 12pt
Line Spacing	1.5 lines (body text) | Single (tables, captions, references)
Paragraph Spacing	6pt before, 6pt after
Alignment	Justified (body) | Centre (title, headings) | Left (bullets)
Page Numbers	Bottom centre, starting from page 1 of main content
Header	Project Title (abbreviated, ≤60 chars) — right-aligned
Total Length	Minimum 10 pages, Maximum 20 pages (excluding references & appendices)
Figures & Tables	Numbered (Figure 1.1,1.2 , Table 1.1, 1.2 according to the chapter in which it appears), with captions below figures and above tables
File Format	Submit as .docx AND .pdf (both required) in the Google classroom folder assigned
Naming Convention	USNs_StudentNames.doc (e.g., 1RV24ISE001_002 _Name1_Name2.pdf)